PRACTICAL NO. 10 – STUDENT MANAGEMENT SYSTEM (LOGIN MODULE)

Aim: To analyze and apply Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on a sample Student Management System (Login Module) and document the testing process, execution, and results.

Test Case ID	Testing Type	Scenario	Test Case Description	Preconditions	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
BB-01	Black-Box Testing	Valid credentials	Verify successful login with correct student credentials	Test account exists; login page available	1. Open login page 2. Enter valid username/password 3. Click Login	student01 / Valid@123	Student is authenticated and redirected to dashboard	Illustrative dashboard represents expected successful flow	Pass (sample)	Positive functional test; SS-01 and SS-04
BB-02	Black-Box Testing	Invalid password	Verify incorrect credentials are rejected	Test account exists	1. Open login page 2. Enter username and wrong password 3. Click Login	student01 / Wrong@123	Error appears; dashboard is not opened	Illustrative invalid-login screen shows error	Pass (sample)	Negative test; SS-02
BB-03	Black-Box Testing	Blank fields	Check required-field validation	Login page available	1. Leave both fields blank 2. Click Login	Username blank; Password blank	Validation appears or login is prevented	Illustrative screen shows required-field message	Pass (sample)	Input validation; SS-03
WB-01	White-Box Testing	Credential comparison branch	Review matching and non-matching credential branches	Login source code/pseudocode available	1. Inspect credential comparison 2. Trace matching branch 3. Trace mismatch branch	Matching and mismatching pairs	Matching branch grants access; mismatch rejects access	Logic path documented in sample test design; not run against source code	Pass (logic review)	Verify source code if live evidence is required
WB-02	White-Box Testing	Empty-input branch	Review missing username/password condition	Validation source code available	1. Inspect empty-field condition 2. Test blank username 3. Test blank password	Username blank / Password blank	Empty-input branch returns validation and does not authenticate	Branch is included in test design	Pass (logic review)	SS-03 shows intended output
WB-03	White-Box Testing	Authentication failure path	Ensure failed login does not create an authenticated session	Authentication logic available	1. Trace failed-auth branch 2. Check session creation 3. Confirm return to login	Incorrect credentials	Access denied; no authenticated session created	Failure behaviour specified; requires real source/runtime verification	Pass (logic review)	Confirm on actual system before claiming code execution
IT-01	Integration Testing	Login and student database	Verify credentials are checked against student records	Login connected to test database	1. Submit registered test credentials 2. Verify lookup response 3. Check student mapping	Registered test credentials	Correct student record is validated	Sample integration flow documented; live DB not connected here	Pass (sample flow)	Live verification requires test database
IT-02	Integration Testing	Authentication and session	Verify successful authentication creates a session	Authentication/session/dashboard integrated	1. Login with valid credentials 2. Check session 3. Open dashboard	Valid student credentials	Session is created and dashboard opens	Dashboard mockup illustrates intended outcome	Pass (sample flow)	SS-04; verify live session on actual system
IT-03	Integration Testing	Login UI and auth response	Verify UI handles authentication rejection	Login UI and auth service integrated	1. Submit invalid credentials 2. Observe UI response 3. Confirm no dashboard access	Invalid credentials	Error shown and user remains unauthenticated	Sample UI displays rejection message	Pass (sample flow)	SS-02
ST-01	System Testing	End-to-end successful login	Test complete login workflow	System and test account available	1. Open system 2. Enter valid credentials 3. Click Login 4. Verify dashboard	Valid credentials	Student reaches correct dashboard	Illustrative flow reaches dashboard screen	Pass (sample flow)	SS-01 and SS-04
ST-02	System Testing	End-to-end failed login	Ensure invalid credentials cannot access protected pages	Login page available	1. Enter invalid credentials 2. Submit 3. Try dashboard URL	Invalid credentials	Login rejected; protected dashboard remains inaccessible	Illustrative error state shown; live access control requires verification	Pass (sample flow)	SS-02
ST-03	System Testing	Logout/session handling	Verify logout ends access to protected pages	Student is logged in	1. Click Logout 2. Try Back button or dashboard URL	Active session	User returns to login and protected content is inaccessible	Illustrative logged-out state; live session invalidation requires verification	Pass (sample plan)	SS-05

Screenshot ID	Screen	Purpose	Included in Word	Evidence note
SS-01	Login Page	Login form with username and password fields.	Yes	Illustrative mockup; not live capture
SS-02	Invalid Login	Error message for incorrect credentials.	Yes	Illustrative mockup; not live capture
SS-03	Empty Input Validation	Required-field message for blank inputs.	Yes	Illustrative mockup; not live capture
SS-04	Student Dashboard	Illustrative destination after successful login.	Yes	Illustrative mockup; not live capture
SS-05	Logout Confirmation	Illustrative logged-out state.	Yes	Illustrative mockup; not live capture

PRACTICAL NO. 10 – NOTES	
Module	Student Management System – Login Module
Sample status	Pass (sample) refers to illustrative UI/test design, not proof of execution on your institution's live application.
White-Box Testing	Review actual source code/pseudocode and branch coverage if live code-based evidence is required.
Integration Testing	A real database/authentication/session environment is required for live integration verification.
Screenshots	Five illustrative screenshots are embedded in the Word report. Replace with live screenshots if required by faculty.
Credentials	Use dummy credentials only; never include a personal password.
Before upload	Add tester name/date and actual test-system URL if available.